Physiology of Testicular Function

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No conflicts of interest to disclose

Learning Objectives

1. Explain the role of gene regulatory proteins (SRY/TDF), Leydig cells, Sertoli cells, and their hormonal products in testicular development and sexual differentiation.
2. Predict the developmental consequences of impaired androgen production or signaling during sexual differentiation.
3. Describe the regulation of testicular function by the hypothalamic-pituitary-gonadal (HPG) axis, including the actions and feedback mechanisms of GnRH, LH, FSH, testosterone, estradiol, and inhibin.

Learning Objectives

4. Compare the actions of testosterone, dihydrotestosterone (DHT), and estradiol in reproductive development, puberty, and adult reproductive function.

5. Explain the stages of spermatogenesis and the roles of Sertoli cells, Leydig cells, and reproductive hormones in regulating sperm production.

6. Explain the neural and vascular mechanisms underlying erection and ejaculation.

Introduction

- Human reproduction involves sperm production by the person with testes, which requires normal development of genital organs, production of androgens, proliferation of the germ cell and differentiation into mature sperm.
- We will explore each of these aspects

Testicular Reproductive Physiology

Clinical aspects:

- A testicular fertility problem is present in 50% cases of couple infertility.
- Androgens are potent steroid hormones that have multiple functions.
- Genetic disorders in the system, such as impaired androgen production or sensitivity, result in infertility and developmental disorders

1. **Genetic program for genital development**
   SRY/TDF, Leydig cells, Sertoli cells, Testosterone, AMH

2. **Androgen physiology:**
   Communications between hypothalamus-pituitary axis, Leydig cell and Sertoli cell through GnRH, LH, FSH, inhibin, testosterone, DHT and estradiol

3. **Sperm production and transportation**

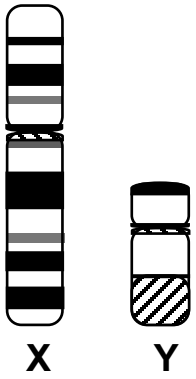
4. **Erection and ejaculation response**

It starts from the Y-chromosome

Human chromosome

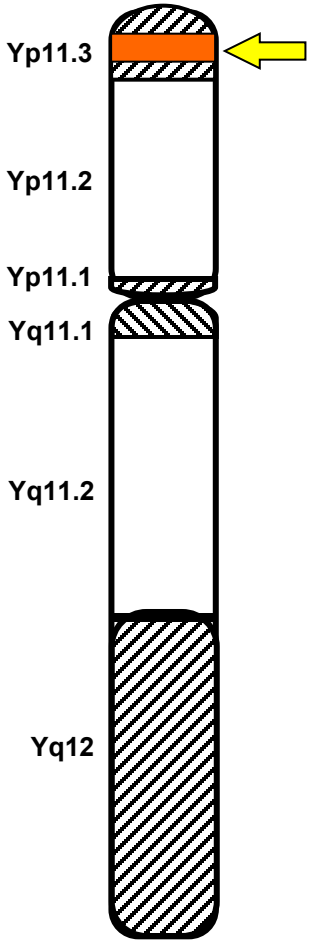
22 pairs of
autosomal chromosomes

Plus, a pair of
Sex chromosomes



SRY/TDF is necessary for testicular development

Y-chromosome



Sex-determining Region Y (SRY)
A gene necessary for *virilization,* the biological development of the phenotypic male body marking the sexual differentiation from phenotypic female.

SRY gene product
= **Testis-Determining Factor (TDF)**

A transcription factor that triggers a series of gene expressions, required for testicular development and virilization

SRY triggers testicular development

Wolffian duct

SRY → Gonads

Mullerian duct

Development of Sertoli cell and Leydig cell

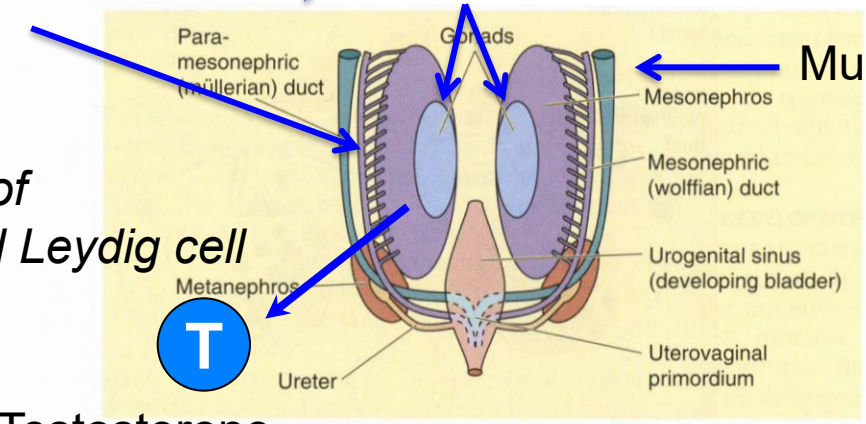
T

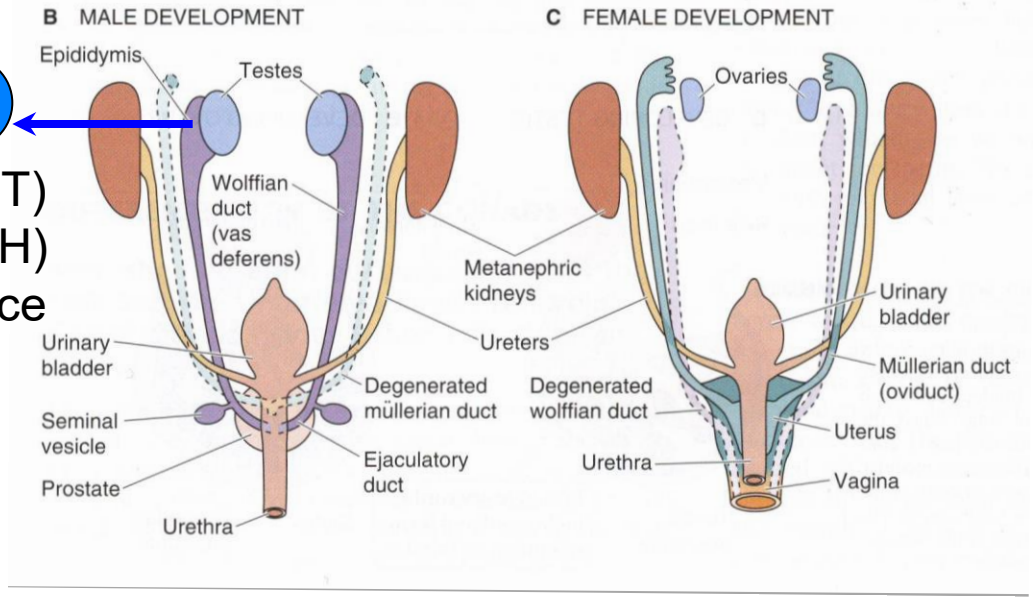
9 weeks after fertilization

Testosterone

More T

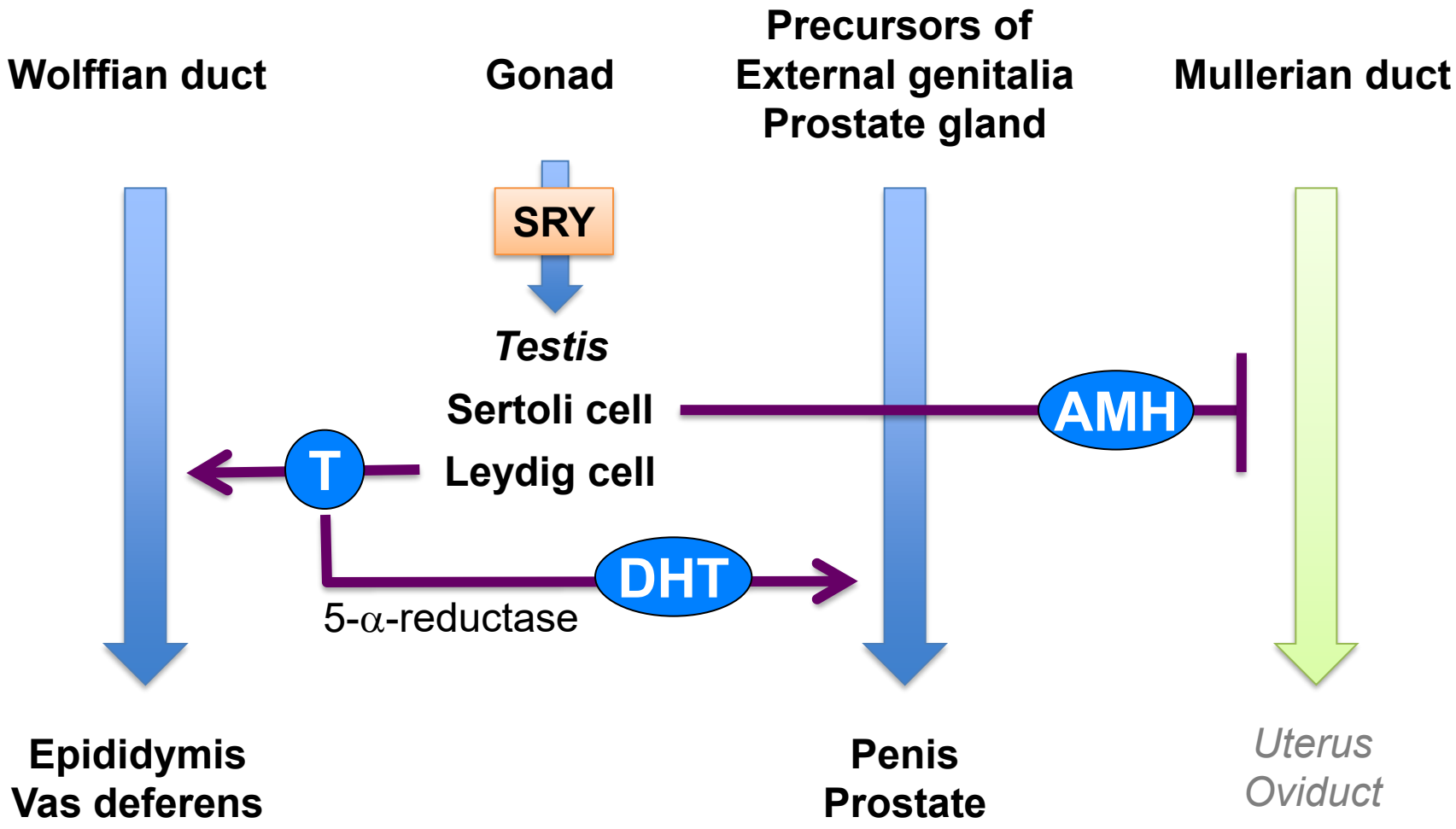
Dihydrotestosterone (DHT)
Antimullerian hormone (AMH)
or mullerian-inhibiting substance





Boron & Boulpaep, Medical Physiology Fig. 52-4

SRY-mediated sexual differentiation



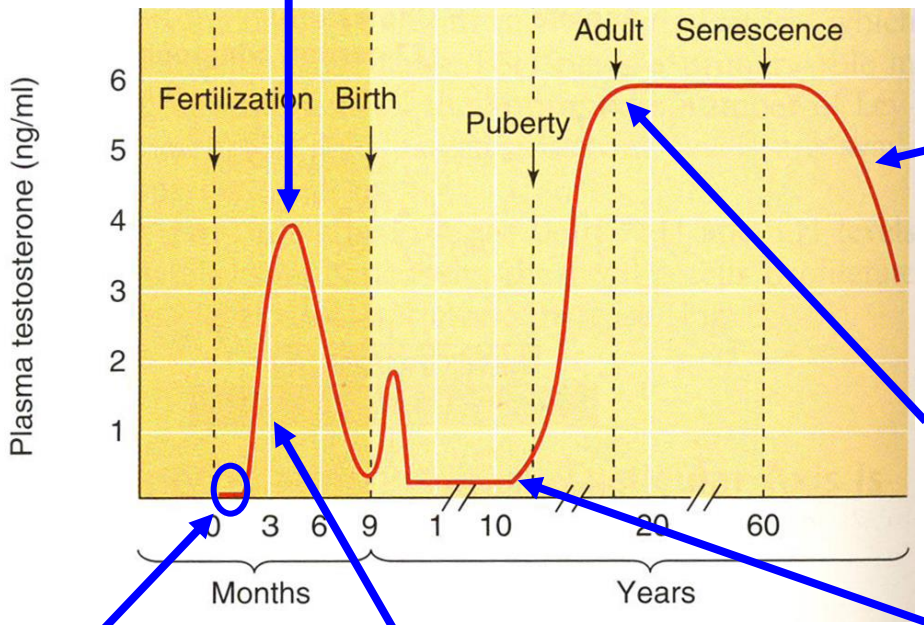
Genotype vs Phenotype in Reproductive Development

Genotype	*Gonad*	*Mullerian duct*	*Wolffian duct*	*Phenotype*
XY, Normal	testis	Regressed	Developed	M
XX, Normal	ovary	Developed	Regressed	F
XO, Turner's syndrome	streak	Developed	Regressed	F
XY w/ Deficiency in SRY	testis*	Developed	Regressed	F
XY w/ Deficiency in testosterone signal	testis	Regressed	Regressed**	M/F
XY w/ Deficiency in AMH signal	testis	Developed	Developed	M
XXY, Klinefelter's syndrome	testis*	Regressed	Developed	M
XX w/ SRY translocation	testis*	Regressed	Developed	M

** Immature testis development, mostly sterile.*

*** Depending on the degree of deficiency, partial Wolffian Duct development may occur.*

Developmental changes in plasma testosterone



Completion of genital system development

Andropause
- **Reduction in GnRH release**
- **Loss of Leydig cell responsiveness**

Negative feedback through hypothalamus

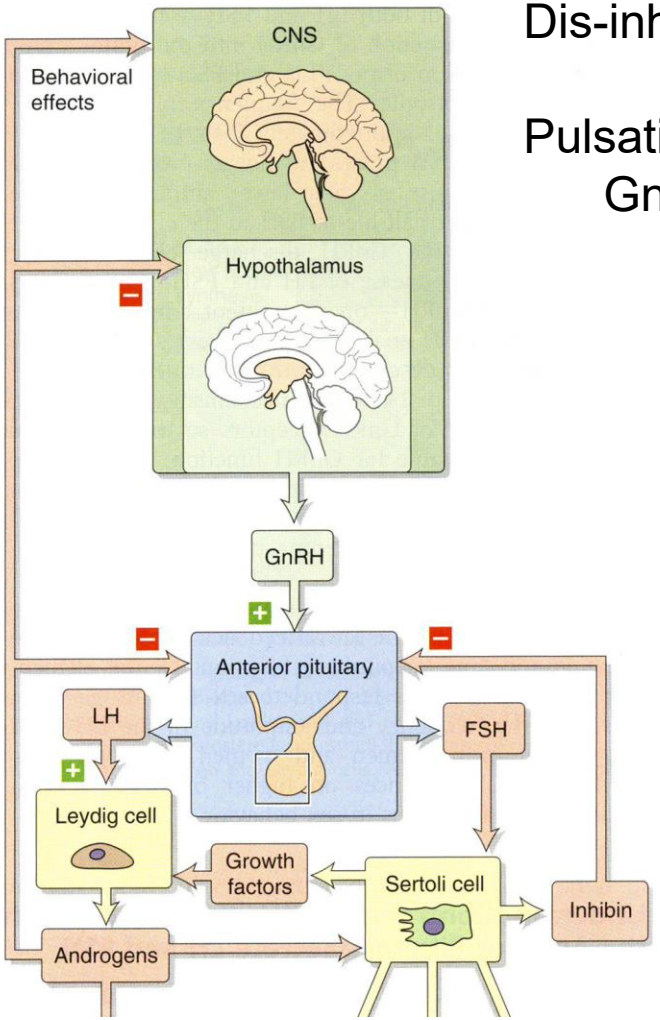
Triggered by hypothalamic-pituitary axis

SRY induces development of Sertoli and Leydig cells (9 weeks)

Growth of Leydig cells

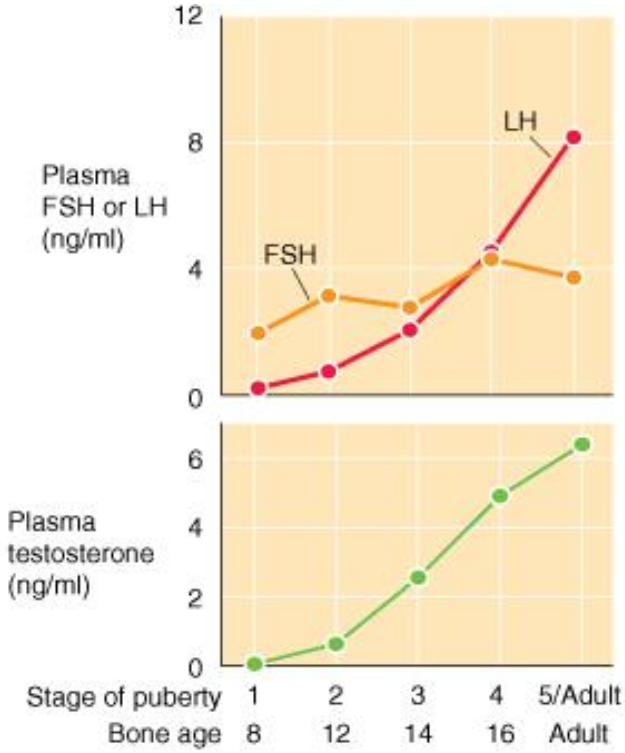
Boron & Boulpaep, Medical Physiology Fig. 53-6

Hypothalamic-pituitary axis and testosterone release



Dis-inhibition of GnRH pulse generator

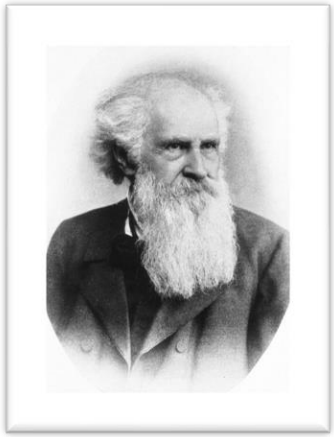
Pulsatile secretion of GnRH from GnRH-secreting neuron (one pulse / 1 hr)



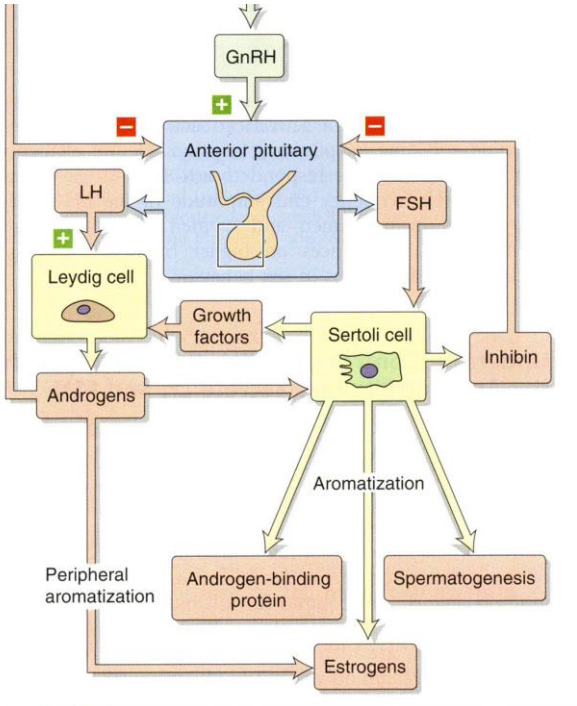
LH (Luteinizing hormone) *FSH (Follicle-stimulating hormone)*

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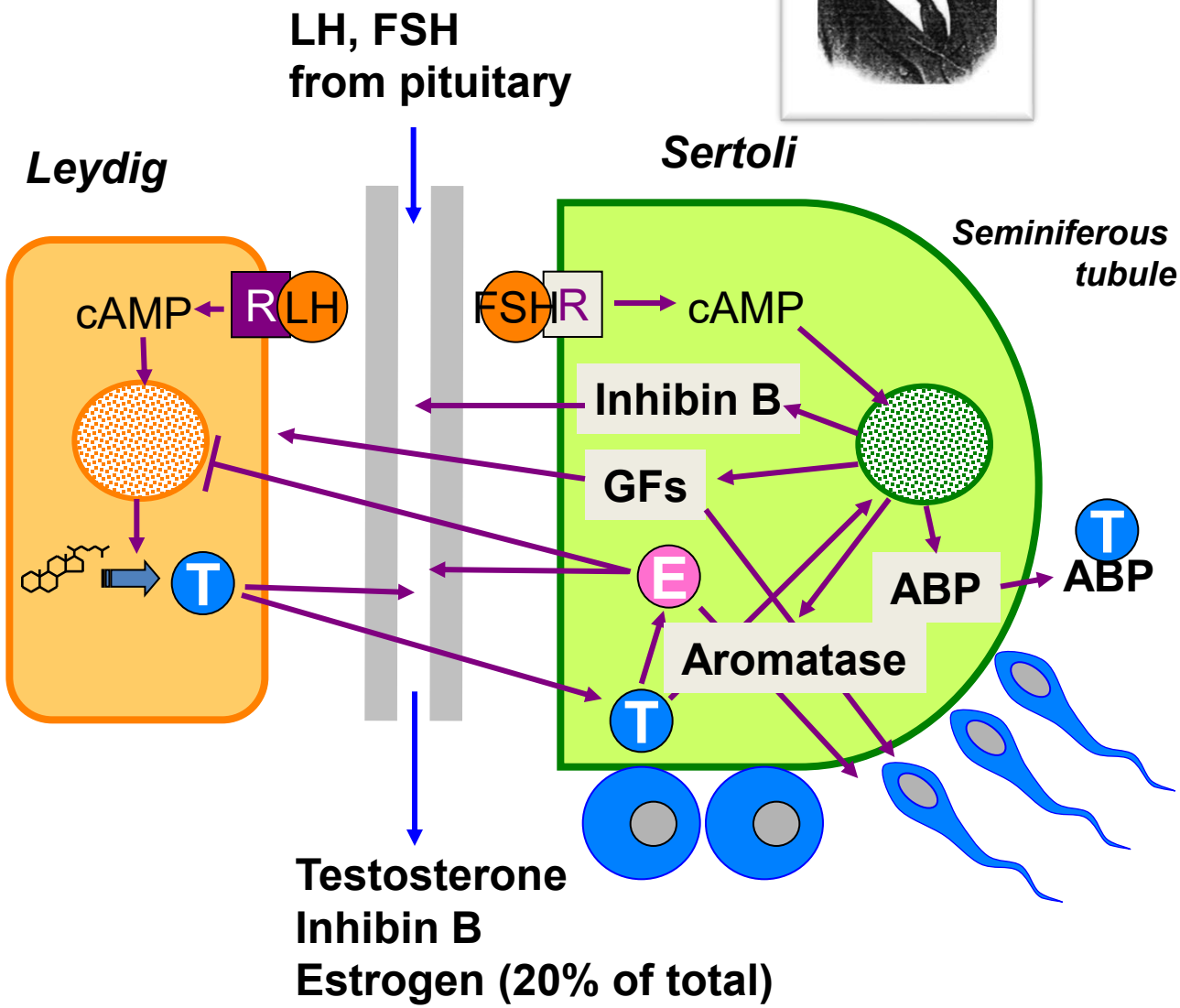
Communication between Leydig and Sertoli cells





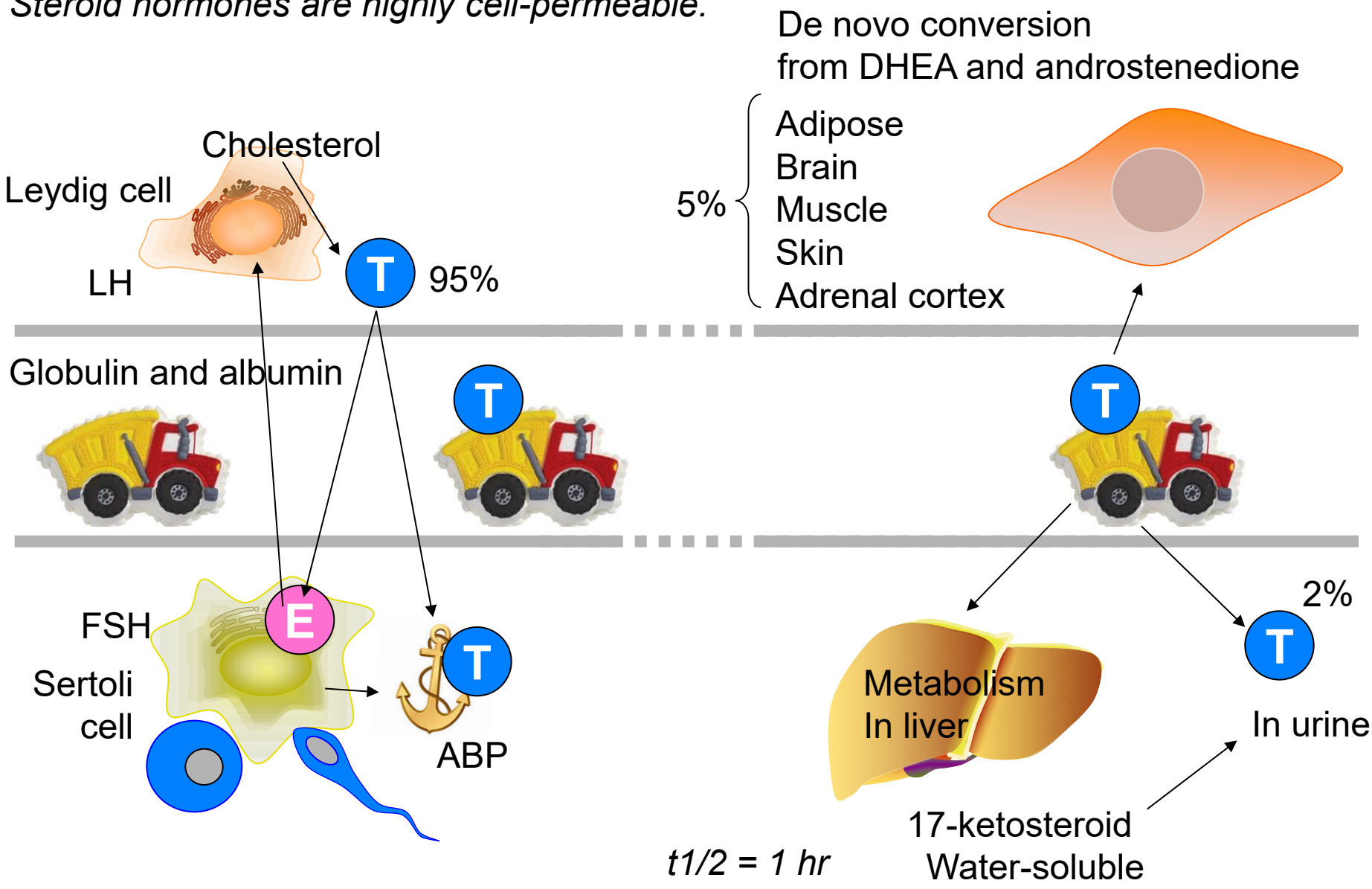


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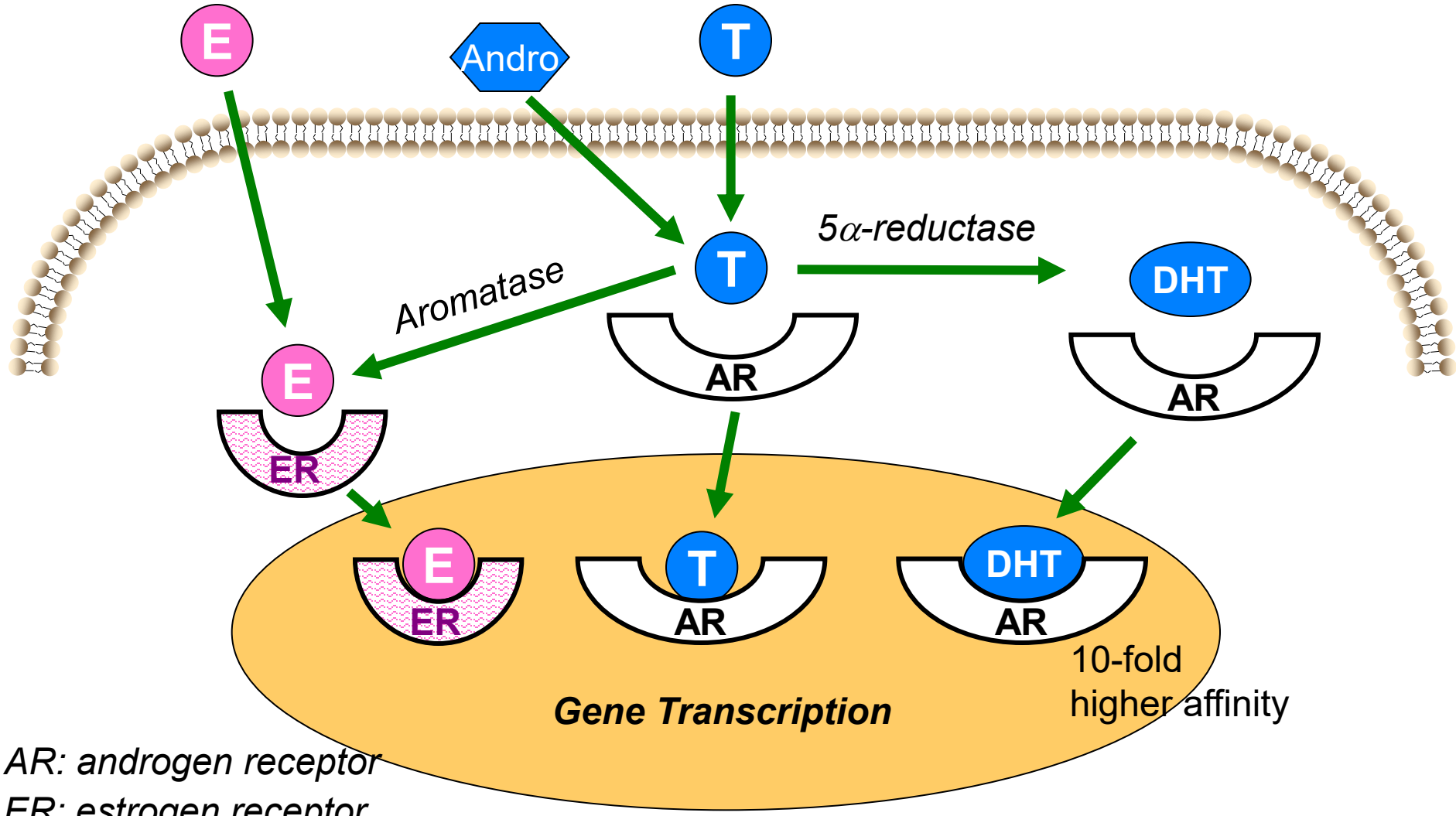


Life of androgens in an XY individual

Steroid hormones are highly cell-permeable.



Cell signaling of androgens at target



AR: androgen receptor

ER: estrogen receptor

Actions of testosterone, DHT and estradiol in the XY individual

Testosterone

- Development of ***internal*** genital tract (Wolffian duct-derived structure)
  epididymis, vas deferens, seminal vesicles
- Pubertal growth of penis and seminal vesicle
- Stimulation of spermatogenesis
- Libido
- Pubertal growth of larynx and deepening of voice
- Pubertal growth spurt: anabolic effects on muscle, bones, and erythropoiesis
- Inhibition of breast development
- Feedback inhibition of GnRH, LH, and FSH release

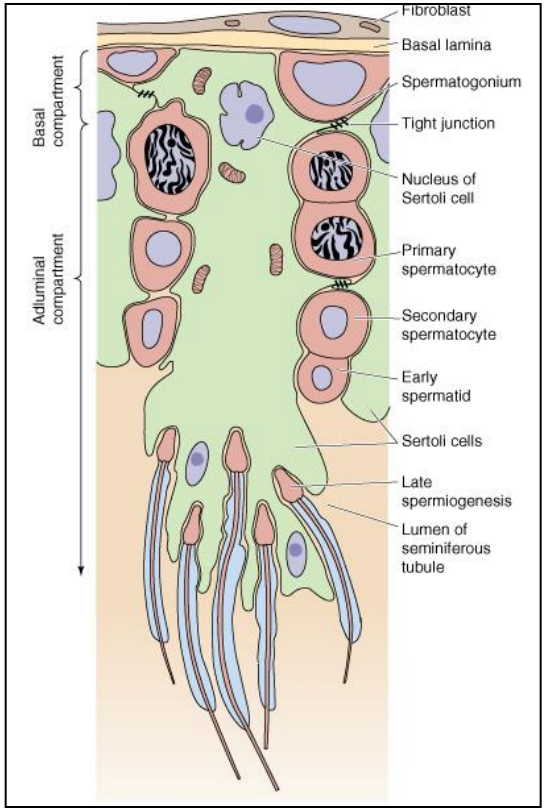
DHT <<< *5α-reductase*

- Development of prostate, penis and scrotum
- Development of pubic and underarm hair, “male pattern” balding
- Activity of sebaceous glands (acne)

Estradiol <<< *Aromatase*

- Epiphyseal growth plate closure (Length of bone), prevention of osteoporosis
- Feedback inhibition of GnRH secretion

Physiology of spermatogenesis



Boron & Boulpaep, Medical Physiology

70 days

Proliferation of spermatogonia, stem cell >>> spermatocyte

Meiosis of spermatocyte >>> spermatid

Spermiogenesis: maturation of spermatid >>> sperm

Spermiation: release from Sertoli cell

further maturation at epididymis (2-4 weeks)

LH, FSH

T

16-day period : Wave of spermatogenesis

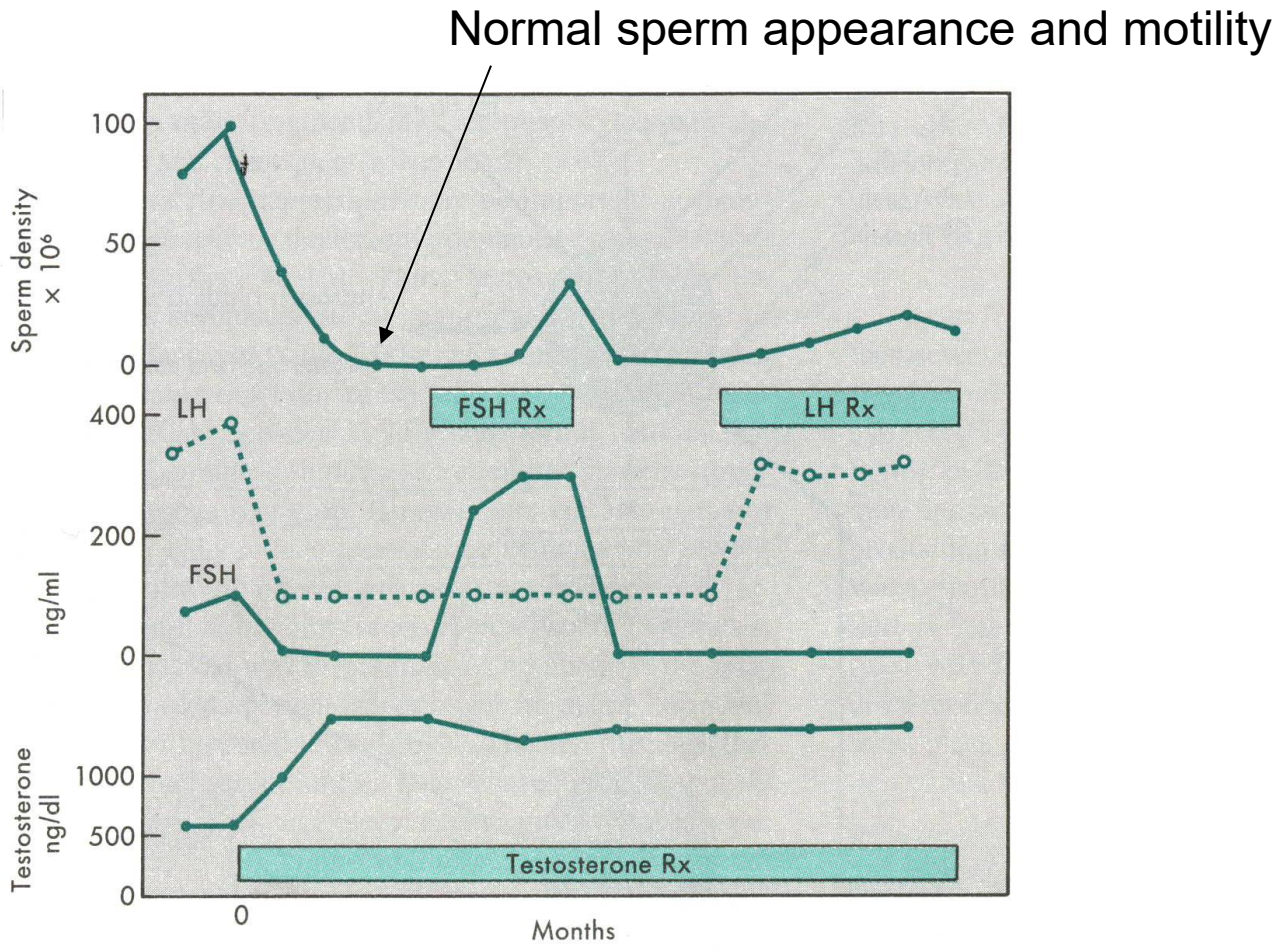
1 spermatogonium >>> 64 sperm cells

>100 million / day

Temperature sensitive:

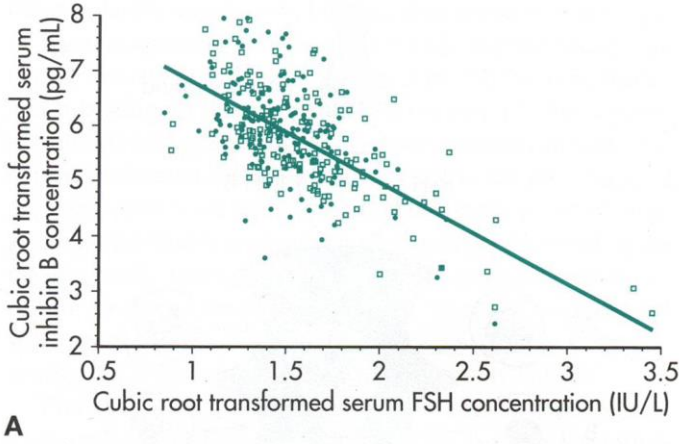
infertility in cryptorchidism (undescended testicle)

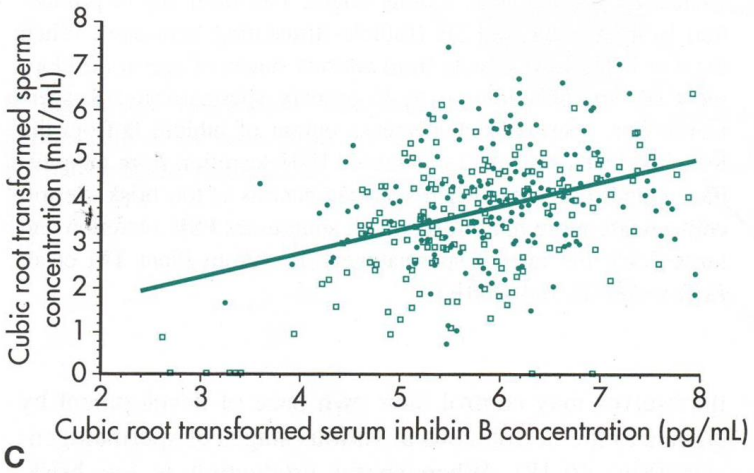
Effects of administration of testosterone, FSH and LH on sperm production

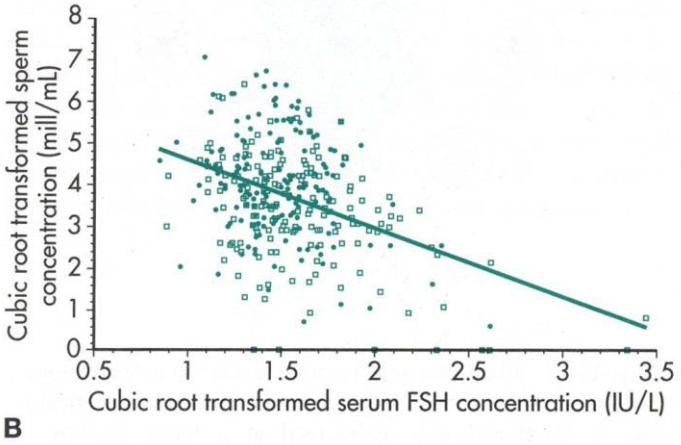


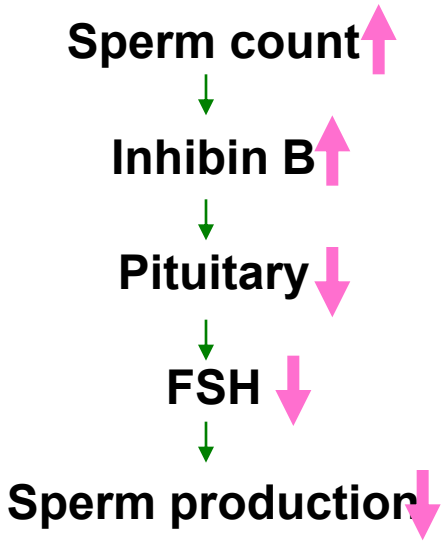
Berne, Levy, Koeppen, Stanton: Physiology, 5th Ed

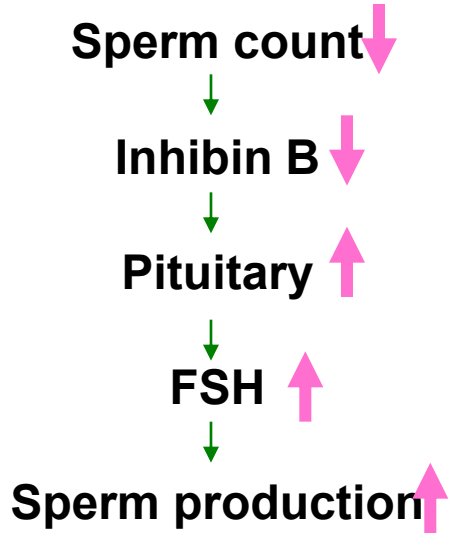
Inhibin B levels correlate with total sperm count











Berne, Levy, Koeppen, Stanton: Physiology, 5th Ed

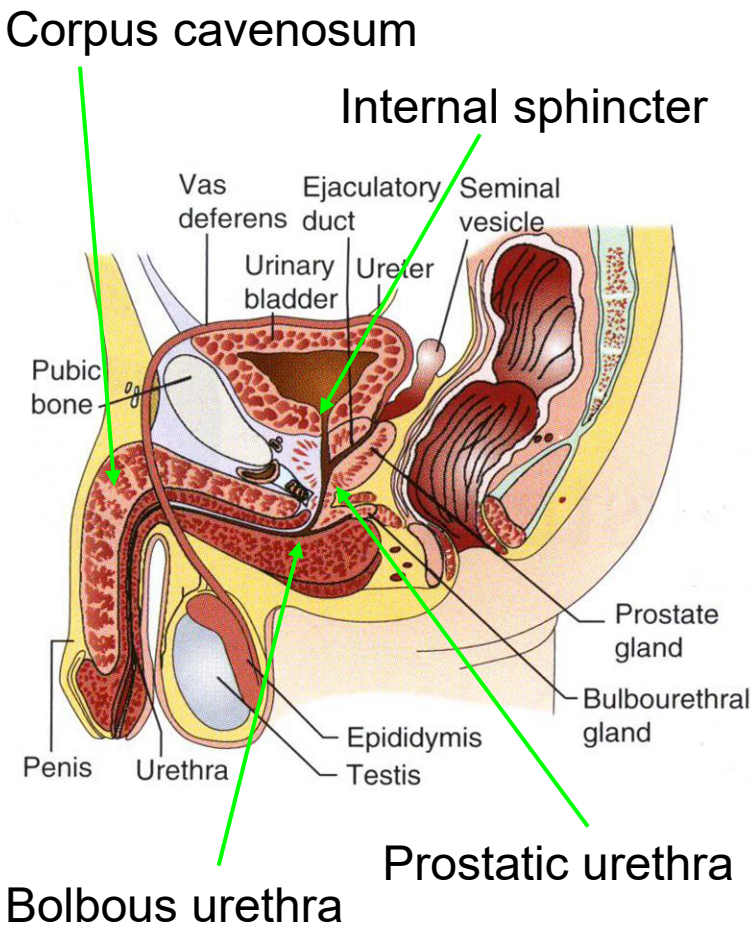
Erection and ejaculation response

1. **Erection**
   Parasympathetic neuron signal (Ach) activates vascular endothelium to produce nitric oxide (NO) at corporeal vasculatures. NO induces vascular smooth muscle relaxation, increases blood flow, and fill blood in corpus cavernosum.

2. **Emission**
   Sympathetic neurons induces peristalic motions of vas deferens, seminal vesicles and prostatic smooth muscles, and transfers seminal fluid to prostatic urethra. Contraction of internal sphincter of bladder blocks ejaculation of sperms into bladder.

3. **Ejaculation**
   Entry of semen to bulbous urethra triggers a reflex reaction, which induces the contraction of the perineal striated muscles to push semen out.

4. **Detumescence**
   Sympathetic nerves induces the contraction of corporeal vasculature.



2 mL x >20 million/mL
= 40-100 mil.
44mm/min = 2.6 m/h
Lifespan of 3-5 days

Testicular Reproductive Physiology

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3. **Sperm production and transportation**

4. **Erection and ejaculation response**

Thank you!